INCH-POUND
MIL-DTL-24643/67B
27 November 2023
SUPERSEDING
MIL-DTL-24643/67A(SH)
w/INT. AMENDMENT 1
11 July 2011
MIL-DTL-24643/67A
28 January 2010

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +221 DEGREES FAHRENHEIT (+105 DEGREES CELSIUS), 1,000 VOLTS, TYPE LS3OW

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Watertight with Circuit Integrity)

First	-	Copper conductor, coated or uncoated, in accordance with ASTM B8 or Navy Standard (see table I for size).
Second	-	Thermoset insulation, extruded or taped (see table I for wall thickness).
Third	-	An optional glass braid.
Fourth	-	An optional covering, unless required (see 3.3.6 of MIL-DTL-24643).
Fifth	-	Three conductors shall be cabled together with a lay in accordance with 3.4.4 of MIL-DTL-24643. Fillers may be used as necessary to form a firm, well-rounded assembly. Standard identification code applied by method 1 or 3, or letter identification code applied by method 5. If identification method 1 is used, outer surface to be printed shall appear black.
Sixth	-	Aluminum foil/polyester tape/aluminum foil, 0.002 inch thick with 45 percent overlap.
Seventh	-	Braided shield of tin-coated copper.
Eighth	-	An optional binder.
Ninth	-	Cross-linked polyolefin jacket. See table I for wall thickness.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

Distribution Statement A. Approved for public release: distribution is unlimited.

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TABLE I. Details.

Military part no. M24643/67	Type and size	Conductor size (AWG/MCM)	Conductor resistance (ohms)	Insulation resistance (megohms)		Insulation wall thickness min. avg. (inch)	Jacket wall thickness min. avg. (inch)	Overall diameter (inches)		Accelerated service loading (amperes)
				Conductor to conductor	Conductor to shield			min.	max.	
-01UO	LS3OW-3	16 (class B)	4.3	500	250	0.018	0.027	0.431	0.463	--
-02UO	LS3OW-4	14 (class B)	2.68	500	250	0.018	0.028	0.449	0.482	--
-03UO	LS3OW-9	10 (class B)	1.08	500	250	0.018	0.036	0.540	0.580	--
-04UO	LS3OW-14	9 (class B)	0.859	500	250	0.018	0.040	0.684	0.733	90
-05UO	LS3OW-23	7 (class B)	0.543	500	250	0.030	0.050	0.769	0.853	115
-06UO	LS3OW-50	3 (class C)	0.21	200	100	0.030	0.050	0.957	1.027	185
-07UO	LS3OW-75	1 (class C)	0.134	175	85	0.035	0.050	1.107	1.187	215
-08UO	LS3OW-100	0 (class D)	0.106	160	80	0.035	0.060	1.231	1.321	295
-09UO	LS3OW-200	0000 (class D)	0.053	125	60	0.050	0.060	1.630	1.748	460
-10UO	LS3OW-300	300 MCM (class D)	0.0377	110	55	0.050	0.075	1.916	2.055	590
-11UO	LS3OW-400	400 (127) Navy STD	0.0273	100	50	0.050	0.075	2.190	2.349	750

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REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	See table I
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor (sizes 3 through 9)	3,000
Conductor to conductor (sizes 14 through 400)	5,000
Conductor to overall shield (sizes 3 through 9)	1,500
Conductor to overall shield (sizes 14 through 400)	2,500
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	See table I
Conductor to overall shield	See table I
Conductor and shield continuity	No failure
Jacket flaws	No failure

Group A:

Visual and dimensional	No failure
Watertightness (see MIL-DTL-24643 for limits of water leakage)	No failure
Crack resistance (applicable to constructions using glass braids with coverings)	No damage
Abrasion resistance (scrapes, min.) (sizes 23 and smaller) (extruded insulation only)	250

Group B:

Thermoset proof test (percent, max.)	
Insulation (extruded insulation only)	50
Jacket (when tested at 392 °F [200 °C])	50
Drip (203±1.8 °F [95±1 °C]) (see MIL-DTL-24643 for limits of drip allowance)	No failure
Cold bending (cable)	No failure

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Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35

Group C:

Physicals (aged) (air oven)	
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Shrinkage	No failure
Heat distortion (percent of unaged, max.)	30
Permanence of printing (insulation – method 1 only) (cycles, min.)	25
Permanence of printing (jacket) (cycles, min.)	125
Cable fill (sealant) removability	No failure
Surface transfer impedance	
Milliohms per meter (max.)	700
Electromagnetic pulse (EMP) response (dB, min.)	60

Group D:

Gas flame (circuit integrity) (1 hour)	No failure
Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Accelerated service (sizes 14 and larger) (see table I for load current)	No failure
Cold working (-4±3.6 °F [-20±2 °C])	Not applicable
Gas flame (circuit integrity) (3 hours)	No failure
Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure

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Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption (extruded insulation only)	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LS3OW (all sizes)	500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-083)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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